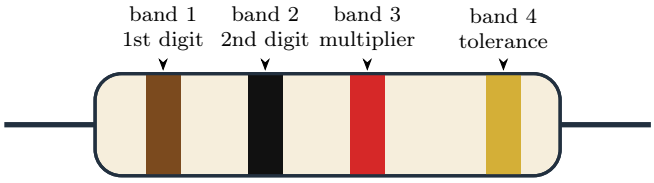
Resistor colour code and the breadboard



brown · black · red · gold = 1 0 × 100 = **1 000 Ω = 1 kΩ, ±5%**

Digits (bands 1, 2) and multiplier (band 3)

Colour	Digit	Multiplier
black	0	×1
brown	1	×10
red	2	×100
orange	3	×1 000 (k)
yellow	4	×10 000
green	5	×100 000
blue	6	×1 000 000 (M)
violet	7	
grey	8	
white	9	

Tolerance (band 4)

gold	±5%
silver	±10%
red	±2%
brown	±1%

Reading a resistor

1. Hold it with the gold or silver band on the **right**.
2. Write the first two colours as a two-digit number.
3. Multiply by the third band.
4. Say the unit: ohms (Ω), kilohms (kΩ) or megohms (MΩ).

Worked example

yellow · violet · red · gold
47 × 100 = 4 700 Ω = 4.7 kΩ, ±5%.
Tolerance means it may really be anywhere from 4 465 to 4 935 Ω.

Units: the same mistake costs marks every year

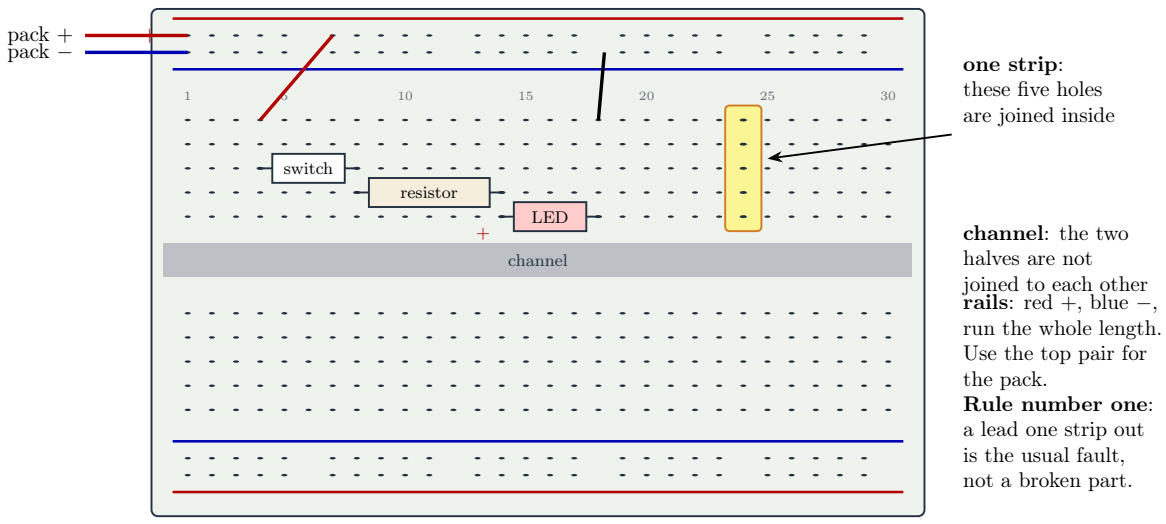
1 kΩ = 1 000 Ω 1 MΩ = 1 000 000 Ω
0.27 kΩ = **270** Ω 4.7 kΩ = **4 700** Ω
Convert to ohms *before* you substitute into a formula.

Standard values you can buy (E12 series), then ×10, ×100, ×1 000 …

10 12 15 18 22 27 33 39 47 56 68 82

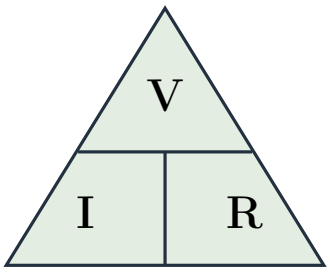
You cannot buy 360 Ω. You pick the **next value up**: 390 Ω (39 × 10). Rounding up keeps the LED current at or below the target.

The 400-point breadboard (84 × 56 mm): 30 strips of five holes on each side of the channel, and four power rails.



Turn over for Ohm's law, measuring and practice.

Ohm's law and the LED resistor



Cover the one you want:

Ohm's law

$$V = I \times R \qquad I = \frac{V}{R} \qquad R = \frac{V}{I}$$

V = voltage in volts (V)
I = current in **amperes** (A)
R = resistance in ohms (Ω)

20 mA = 0.020 A. Divide milliamps by 1 000 before you calculate.

The resistor for an LED: four steps

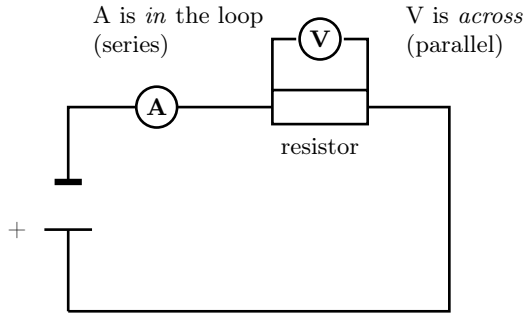
1. Write what you know: supply V_S, the LED's forward voltage V_{LED}, and the current you want I (in amps).
2. The resistor must take the voltage the LED does not: $R = \dfrac{V_S - V_{LED}}{I}$
3. Substitute and calculate, with units.
4. Choose the **next standard value up** from the E12 list, then check the real current $I = \dfrac{V_S - V_{LED}}{R}$.

Worked example: two 18650 cells (7.4 V), an LED of 2.0 V that needs 8 mA

$R = \dfrac{7.4\text{ V} - 2.0\text{ V}}{0.008\text{ A}} = \dfrac{5.4\text{ V}}{0.008\text{ A}} = 675\ \Omega$ → the next value up is **680 Ω**.

Check: $I = \dfrac{5.4\text{ V}}{680\ \Omega} = 7.9$ mA, just below 8 mA. A fully charged pack is 8.4 V and gives $\dfrac{6.4}{680} = 9.4$ mA, still inside the 20% allowed.

Measuring with the multimeter



At the clinic desk

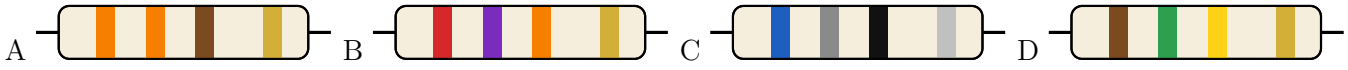
1. **Resistance:** resistor out of the circuit, dial on Ω.
2. **Voltage:** meter *across* the resistor, V DC.
3. **Current:** open the circuit and put the meter *in* it, mA range.
4. Compare $V \div I$ with the number you read from the colours.

Never put the meter on the current range across the battery.

Rules for the 7.4 V pack (two 18650 cells)

Build with the pack unplugged; connect it last and disconnect it first. Two cells can deliver far more current than a 9 V battery if the rails touch: never lay a bare lead across both rails, and tell your teacher at once if anything gets warm.

Try it in your head (answers upside down at the bottom)



A: 330 Ω B: 27 kΩ C: 68 Ω, ±10% D: 150 kΩ

Exam habit: write the formula, substitute with units, give the answer with its unit, then state the standard value you would use.